03. Component Interactions & Sequence Flows

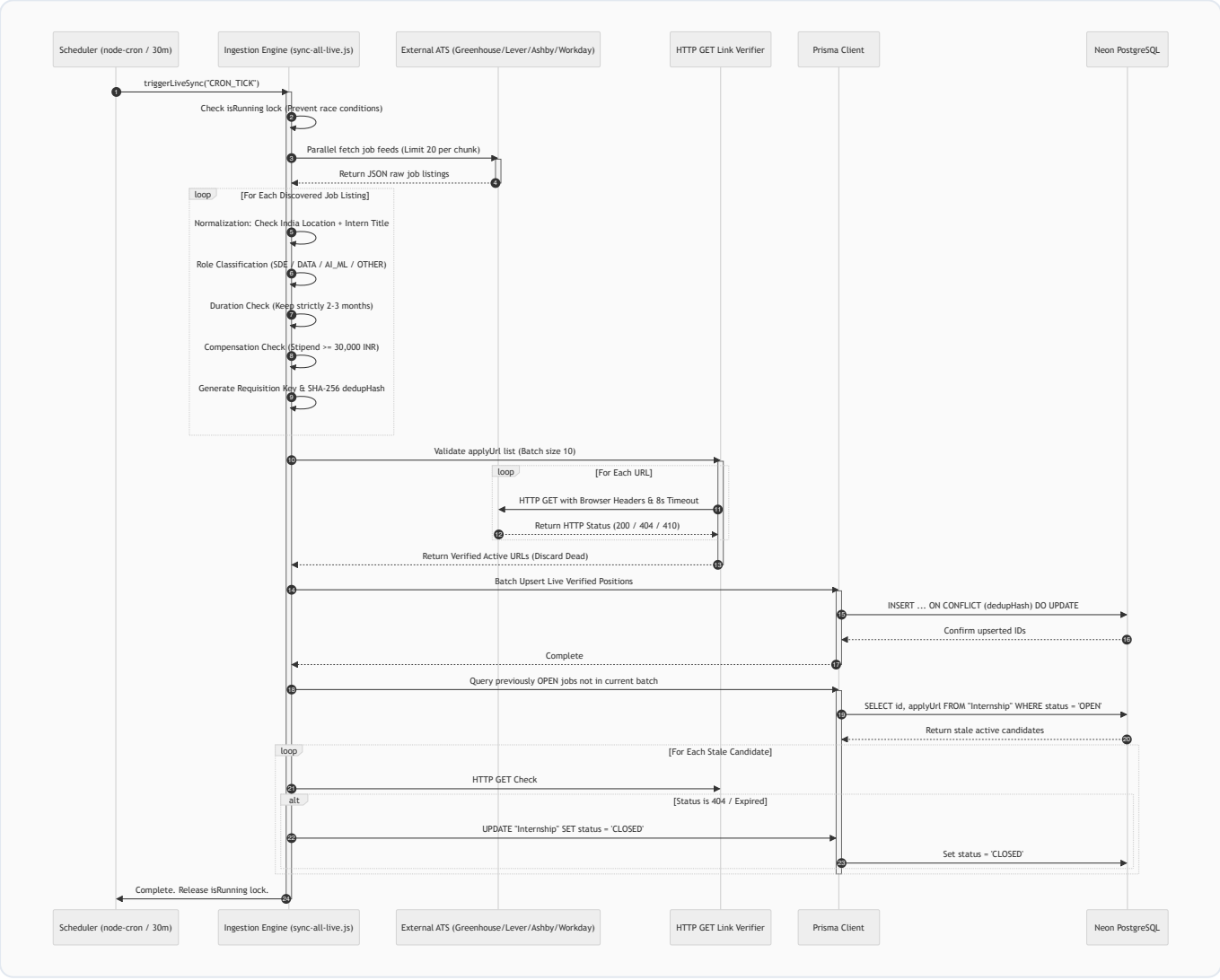
1. System Interaction Overview

This document outlines the detailed sequence flows and component interactions across the Internship Tracker application, from client-side UI actions to backend routing, database queries, and external ATS network calls.

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2. Sequence Diagram 1: Automated Ingestion & Live Link Verification Flow

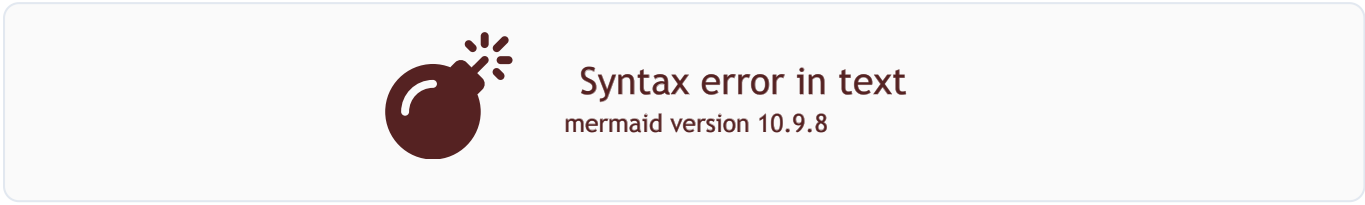
This sequence demonstrates how the system autonomously discovers new internships, verifies them over HTTP, deduplicates against existing records, and updates PostgreSQL.



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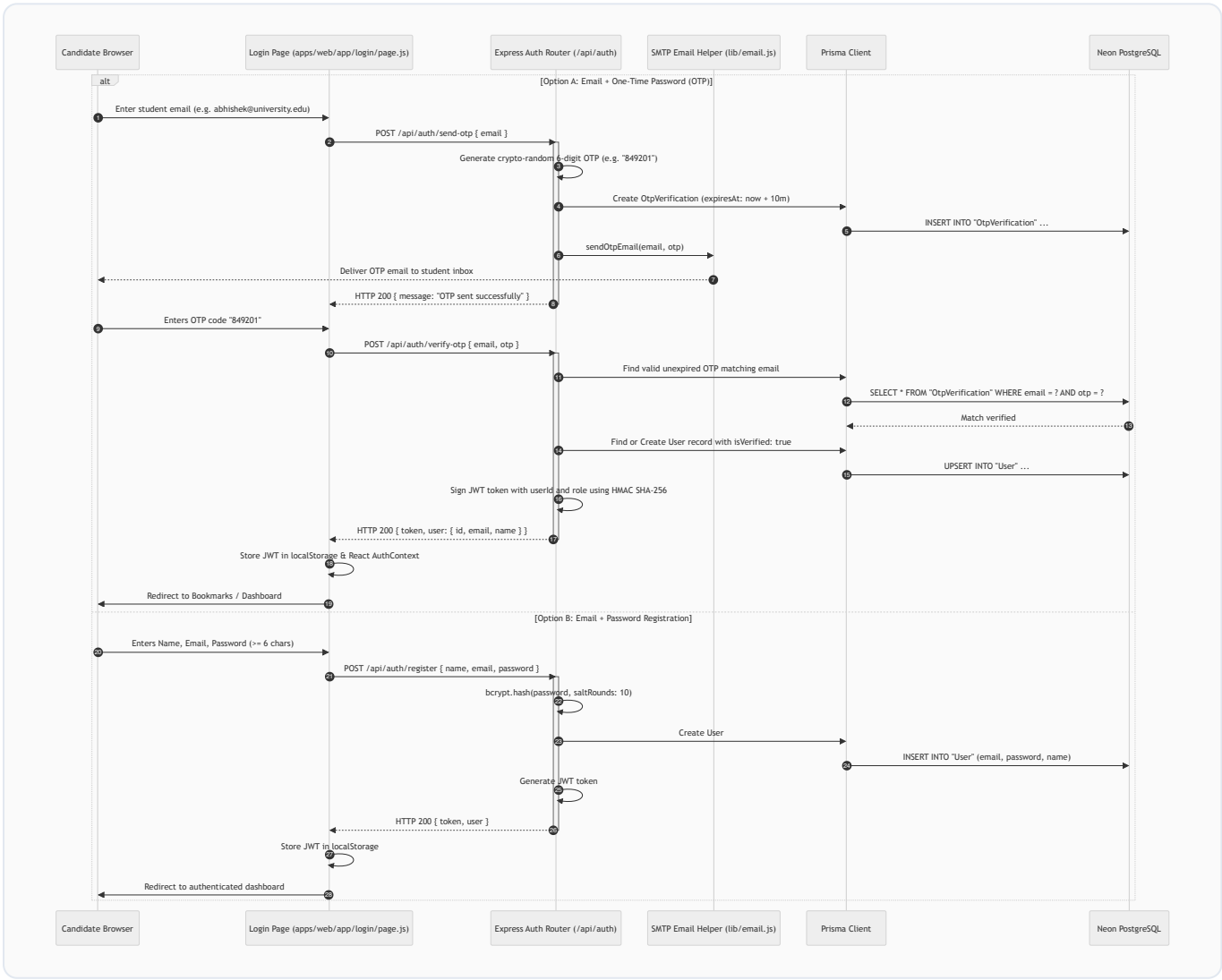
3. Sequence Diagram 2: User Search, Multi-Facet Filtering & Pagination Flow

This sequence demonstrates how a student interacts with the web application to search and filter internships.



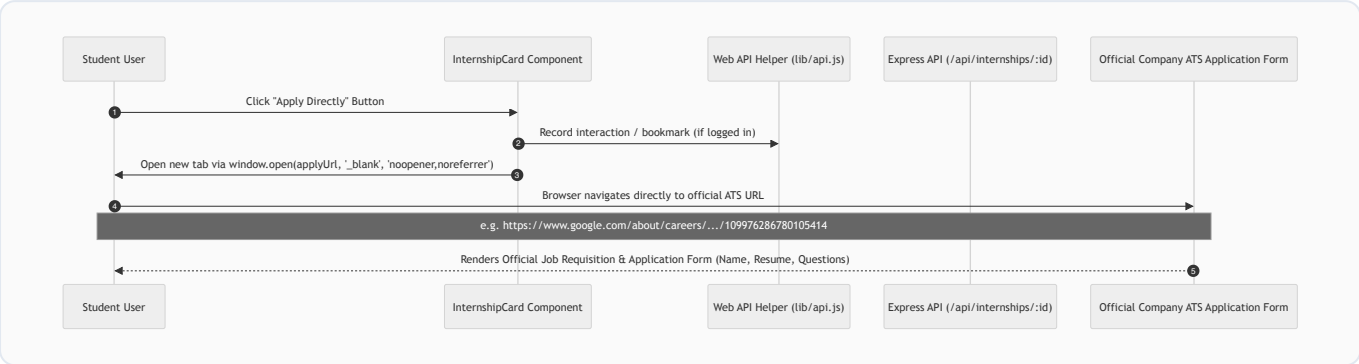
4. Sequence Diagram 3: User Authentication & JWT Session Flow

This sequence demonstrates the passwordless and password-based authentication system with OTP verification.



5. Sequence Diagram 4: Direct Application Redirection & Tracking

This sequence illustrates what occurs when a student clicks **"Apply Directly"**:



6. Frontend Component Architecture & Tree

The frontend is structured in the Next.js 14 App Router pattern:

```
apps/web/
├── app/
│   ├── layout.js                # Global HTML Root, Meta Tags, Navbar, Footer, Providers
│   ├── providers.js             # Theme Provider, Auth Context, TanStack Query Client
│   ├── page.js                  # Main Discovery Dashboard (Hero, Stats, Feed, Pagination)
│   ├── browse/page.js           # Alternate clean browsing view
│   ├── companies/
│   │   ├── page.js              # Alphabetical & Sector Grid of 210+ Tracked Companies
│   │   └── [slug]/page.js       # Company Dossier: HQ, Sector, Active Openings List
│   ├── internships/
│   │   └── [id]/page.js         # Deep Dive Role Spec: Skills, Timeline, Markdown Summary
│   ├── saved/page.js            # User's Bookmarked Internships (Requires Auth)
│   └── login/page.js            # Dual-mode Auth: OTP verification + Password login
│
└── components/
    ├── Navbar.js                # Navigation bar, mobile drawer, auth status pill
    ├── FilterPanel.js           # Multi-select filters: Role, Location, Stipend, Duration, Degree
    ├── SearchBar.js             # Debounced search input + instant submission handler
    ├── InternshipCard.js        # Core listing card: stipend badge, duration, company logo, apply C
    ├── InternshipDetailClient.js # Rich details modal/drawer with full spec rendering
    ├── CompanyInternshipsList.js # Embeddable list of openings on company detail page
    ├── Badge.js                 # Standardized color-coded tags (SDE=Blue, AI=Purple, Data=Green)
    ├── Skeleton.js              # Animated placeholder skeletons for loading states
    ├── Pagination.js            # Responsive page navigation with ellipsis handling
    └── EmptyState.js            # Friendly UI state when no filters match
```

7. Backend Middleware Pipeline

Every HTTP request entering Express is processed in a sequential pipeline before reaching business logic:

1. **`cors()`**: Whitelists frontend domain (`localhost:3000` or production domain) while enabling credential sharing.
2. **`helmet()`**: Injects HTTP security headers (`X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, `Strict-Transport-Security`).
3. **`morgan('dev')`**: Real-time HTTP request logging with method, path, response status, and millisecond latency.
4. **`compression()`**: Gzip/Brotli response compression, reducing payload size by up to 70% for large JSON lists.
5. **`express.json()`**: Parses incoming JSON request bodies with payload size safety caps.
6. **`authLimiter`**: Enforces IP-based rate limiting (100 requests per 15 minutes) on sensitive authentication endpoints.
7. **`requireAuth`**: Validates `Authorization: Bearer <token>`, decodes JWT claims, and attaches `req.user`.
8. **`errorHandler`**: Catch-all global error handling middleware that captures unexpected exceptions, formats standard JSON error payloads, and logs stack traces.